

AWS, Google Cloud and Microsoft. These are commodity infrastructure services and differentiating services.

The infrastructure services include virtual machines, storage volumes, network capabilities and even undifferentiated database services. These are the services that are relatively similar across cloud providers. Therefore, you should evaluate them on the basis of a straightforward price versus performance trade-off. Performance criteria are not limited to actual computational performance but also include geographic location (when it matters for latency or regulatory reasons), availability guarantees, billing granularity and other relevant attributes.

The harder decision is how, when and where to make use of the differentiating service capabilities. These are the services that are unique to each cloud provider, including differentiated machine learning, IoT (Internet of Things) device management, Big Data and other more specific functionality. Selecting to use these types of services can significantly speed up the development of your overall application architecture, but they come with the potential downside of forcing a long-term cloud provider selection based on capability.

Enterprise customers need to find the right balance between these considerations. But they first need to look at what their actual needs are—if you are deploying a modern application or container platform on top of the cloud providers' infrastructure services, you are likely to want to focus on the price versus performance balance as a primary decision point. Then you can cautiously decide to use the differentiating services.

Also, it is not necessarily a decision of which single cloud provider you will use. If your organisation has either a sufficiently advanced operations team or a sufficiently complex set of requirements, you can choose to use multiple providers for what they are best fit for.

Q Do you believe a containerised solution like Cloud Foundry is vital for enterprises moving towards digital transformation?

I believe that digital transformation is fundamentally about changing the nature of an organisation to more readily embrace the use of software (and technology, in general) as a strategic asset. It's a fundamental shift in thinking from IT as a 'cost centre' to IT as a business driver. What matters the most is how an organisation structures its efforts, and how it makes the shift to a product-centric mindset to manage its technology projects.

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That said, Cloud Foundry is increasingly playing a major role as the platform on which organisations restructure their efforts. In many ways, it serves as a 'forcing-function' to help inspire the changes required in an organisation outside of the platform itself. When you take away the technical obstacles to delivering software quickly, it becomes very obvious where the more systemic issues are in your organisation. This is an opportunity.

Q What's your opinion on the concept of serverless computing in the enterprise space?

I prefer the term 'event driven' or 'functions-as-a-service' because the notion of 'serverless' is either completely false or not descriptive enough. There is always a server, or

more likely many servers, as a part of a compute service. Capabilities like AWS Lambda are better described as 'event driven' platforms.

We are early in the evolution and adoption of this developer abstraction. All the large-scale cloud providers are offering event-driven services, like AWS's Lambda. Nevertheless, any new abstraction that is going to bloom in the market needs to have a very active period of discovery by early adopters to drive the creation of the programming frameworks and best practices that are necessary, before it can truly take off within the enterprise context. I believe we are in the early stages of that necessary 'Cambrian explosion'.

Q Is there any plan to expand Cloud Foundry into that model of 'event-driven' computing?

It is quite likely. As with all community driven open source projects, our plans emerge from the collective ideas and actions of our technical community. This makes it impossible to say with certainty what will emerge, outside of the documented and agreed upon roadmaps. However, there have been several proof-of-concepts that have demonstrated how the Cloud Foundry Runtime is well prepared to extend itself into that abstraction type.

Q Lastly, how do you foresee the mix of containers and the cloud?

Containers are, without a doubt, growing in usage. They are being used locally on developer machines, within corporate data centres and within public cloud providers.

The public cloud, on the other hand, is undergoing a massive wave of adoption at the moment. This is not just 'infrastructure services'. Public clouds are offering services that span infrastructure offerings, Platform-as-a-Service, Functions-as-a-Service and Software-as-a-Service. 